

Chapter 2: The DAB (Dual Active Bridge) Topology

Modeling and Optimization of Bidirectional DAB DC–DC Converter Topologies

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Study Notes by Catastrophy_ame

1: Pseudo-Resonant Full Bridge

Divan and De Doncker proposed the *Pseudo-Resonant Full Bridge* DC–DC converter in 1987. Krismer’s thesis does not devote a dedicated section to this circuit, but it is the starting point of the entire derivation chain, so we begin here.

Circuit Structure and the Meaning of “Pseudo-Resonant”

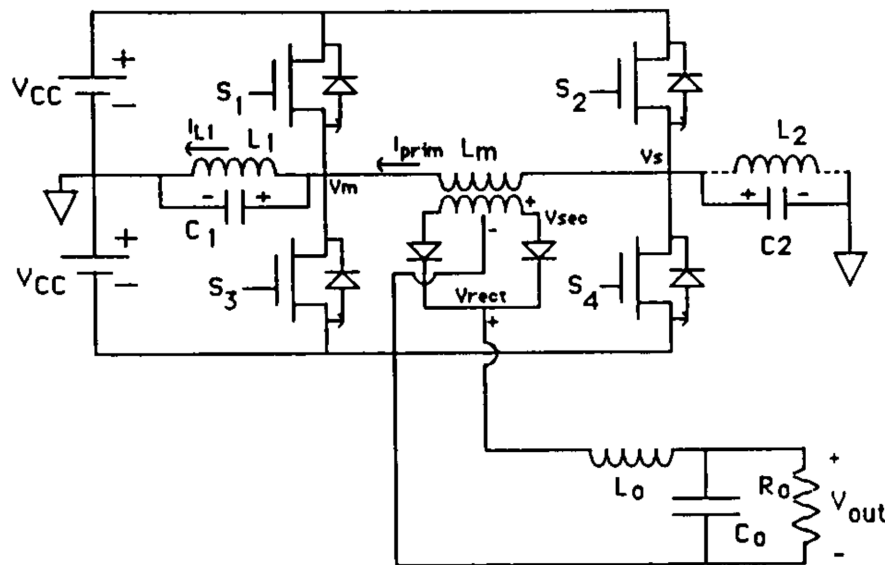


Figure 1: Pseudo-resonant full bridge topology

Characteristics of the pseudo-resonant full bridge:

- Primary side: full bridge (4 active switches) + resonant inductor L_1 + resonant capacitor C_1
- Transformer: HF transformer (center-tapped secondary rectification)
- Secondary side: diode rectifier + DC-side filter inductor L_o + filter capacitor C_o

What “pseudo-resonant” means

- **A true resonant converter** (e.g. the SRC): L and C resonate continuously throughout the switching period, the current is sinusoidal, and power is controlled by frequency modulation (FM).
- **A pseudo-resonant converter**: resonance occurs only *during the switching commutation instants*, and is used to softly transfer the switch voltage/current across the transition. For the rest of the period the circuit behaves like an ordinary square-wave PWM converter.

The Problem: Leakage Inductance vs. Diode Reverse Recovery

The engineering problem of the pseudo-resonant full bridge is that **the interaction between the transformer leakage inductance L_l and the reverse recovery of the secondary-side diodes cannot be reconciled.**

Failure chain

1. The primary square wave reverses polarity $\Rightarrow$ energy $\frac{1}{2}L_l I^2$ is already stored in the leakage inductance L_l .
2. At the same moment a secondary diode is in reverse recovery $\Rightarrow$ during reverse recovery the diode behaves like a short circuit.
3. The current in L_l is forcibly interrupted $\Rightarrow$ a voltage spike $V_{\text{spike}} = L_l \cdot di/dt$ appears across the primary-side devices.
4. The reverse-recovery peak current flows through the switches $\Rightarrow$ this turns directly into switching loss.
5. The higher the switching frequency, the more frequent the L_l /reverse-recovery events $\Rightarrow$ the loss becomes unacceptable.
6. **Result: the switching frequency is capped at some ceiling, and a power-density ceiling appears.**

The fix: PSFB

Move the DC-side filter inductor L_o over to the AC side, so that it is placed in series with the transformer leakage inductance L_l .

What does this move change?

- **Before:** L_l is a parasitic on the AC side and L_o is a filter on the DC side; the two are independent. When L_l undergoes reverse recovery, its energy has nowhere to go $\Rightarrow$ loss.
- **After:** L_o is placed *in series with L_l in the AC loop*, merging the two into a single $(L_l + L_o)$. When a diode reverse-recovers, the current in this “enlarged” inductor **naturally commutates from the original diode to the opposite diode**; the stored energy is not lost and is instead delivered to the load.

This is exactly the statement in the thesis: the energy stored in $(L_l + L_o)$ during reverse recovery is now delivered to the load in a lossless manner rather than being dissipated.

At the same time, since L_o is now large enough, the original resonant inductor L_1 is no longer needed — **phase-shift control** alone is enough to accomplish all regulation + soft switching at a *fixed frequency*. The scheme changes from FM control to PWM phase-shift control.

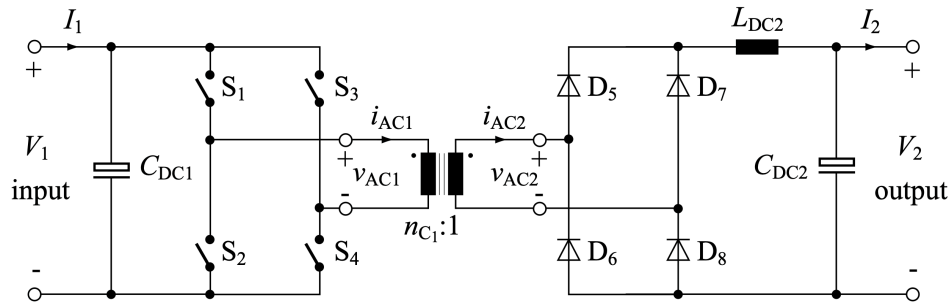
This step turns the pseudo-resonant full bridge into the **Phase-Shift Full Bridge (PSFB)**.

2: Phase-Shift Full Bridge (PSFB)

Circuit Topology

Core components of the PSFB:

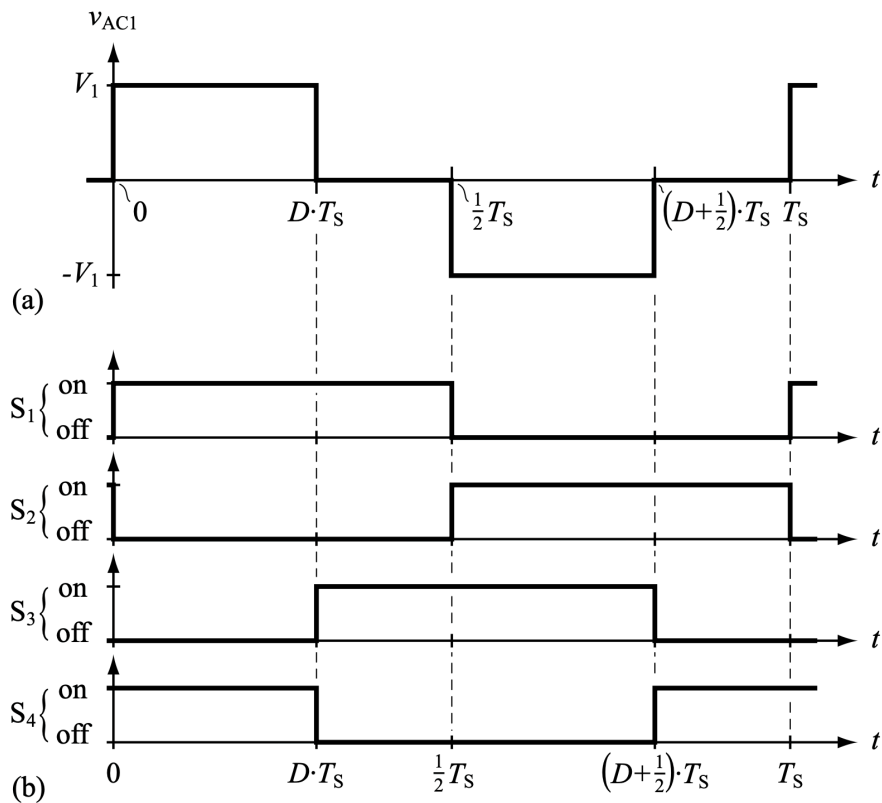
- **Primary side:** 4 active switches S_1 – S_4 forming a full bridge
- **Transformer:** HF transformer, turns ratio $n_{C_1} : 1$
- **Secondary side:** 4 diodes D_5 – D_8 forming a (passive) rectifier bridge
- **Filter:** DC output inductor L_{DC2} + filter capacitor C_{DC2}


 Figure 2: Unidirectional full bridge DC–DC converter with output inductor L_{DC2}

This is a **unidirectional buck-derived** converter — the primary side inverts actively, the secondary side rectifies passively, and power can only flow from port 1 to port 2, because the right-hand side is made of passive components.

The problem: if the switches simply conduct in alternating diagonal pairs, there is no regulation knob. v_{AC1} is always a full-amplitude square wave, the delivered power is always the maximum the circuit can produce, and there is no way to regulate the voltage or handle a changing load. That is unacceptable for industrial applications.

Phase-Shift Control


 Figure 3: (a) Transformer voltage $v_{AC1}(t)$ generated by the inverter stage of the DC–DC converter; (b) the respective switching states

The idea of phase-shift control: split the 4 switches into **two half-bridges**, and make the right

half-bridge start $D \cdot T_S$ later than the left half-bridge.

How the Three-Level v_{AC1} Waveform Is Produced

Plotting the voltage difference between the two half-bridge midpoints ($v_{AC1} = V_A - V_B$, where A is the left midpoint and B is the right midpoint):

Time interval	V_A	V_B	v_{AC1}	Switches ON
$0 \rightarrow D \cdot T_S$	V_1	0	$+V_1$	S_1, S_4 (diagonal)
$D \cdot T_S \rightarrow T_S/2$	V_1	V_1	0	S_1, S_3 (top short)
$T_S/2 \rightarrow (D + 1/2)T_S$	0	V_1	$-V_1$	S_2, S_3 (diagonal)
$(D + 1/2)T_S \rightarrow T_S$	0	0	0	S_2, S_4 (bottom short)

The diagonal pair is still the core mechanism that produces $\pm V_1$; it is simply that two **freewheeling phases** (top short + bottom short) are inserted in between, so that v_{AC1} exhibits a zero plateau.

- Node A is connected to the top rail through the low-resistance channel of $S_1 \Rightarrow V_A \approx V_1$.
- Node B is connected to the top rail through the low-resistance channel of $S_3 \Rightarrow V_B \approx V_1$.
- $v_{AC1} = V_A - V_B \approx 0$.

A and B are both shorted to **the same rail (equipotential)**, so there is no voltage difference across the primary of the transformer and $v_{AC1} = 0$.

Bidirectionality of the MOSFET channel

The MOSFET channel is the conductive layer induced at the surface of the silicon substrate once a positive voltage is applied to the gate. It is essentially **an induced low-resistance resistor** with no preferred direction — current can enter from either end. Therefore:

- ON state: current can flow drain $\rightarrow$ source (forward), voltage drop $= I \cdot R_{DS(on)}$.
- ON state: current can also flow source $\rightarrow$ drain (reverse), voltage drop $= I \cdot R_{DS(on)}$.
- OFF state: only then does the body diode (antiparallel diode) take over — unidirectional conduction, voltage drop $\approx V_F = 0.7 \text{ V}$.

Power Formula of the PSFB

Referring v_{AC1} through the transformer to the secondary side ($v_{AC2} = v_{AC1}/n_{C1}$) and taking the average after rectification through L_{DC2} gives:

$$V_2 = 2D \cdot \frac{V_1}{n_{C1}} \quad (1)$$

Within each period, **the total fraction of time v_{AC1} has voltage applied is $2D$** (in each half-period the “ $+V_1$ duration” $= D \cdot T_S$, and in the opposite half-period the “ $-V_1$ duration” is also $D \cdot T_S$; after rectification both contribute positive pulses). This pulse train is averaged by L_{DC2} , giving the average output $V_2 = 2D \cdot V_1/n_{C1}$.

The fundamental property of the PSFB: it can only step down

Since $D \leq 0.5$ (buck-derived operation), Eq. (1) gives $V_2 \leq V_1/n_{C1}$. **V_2 can never exceed V_1/n_{C1}** . This limitation directly leads to the “turns-ratio contradiction” in the next section.

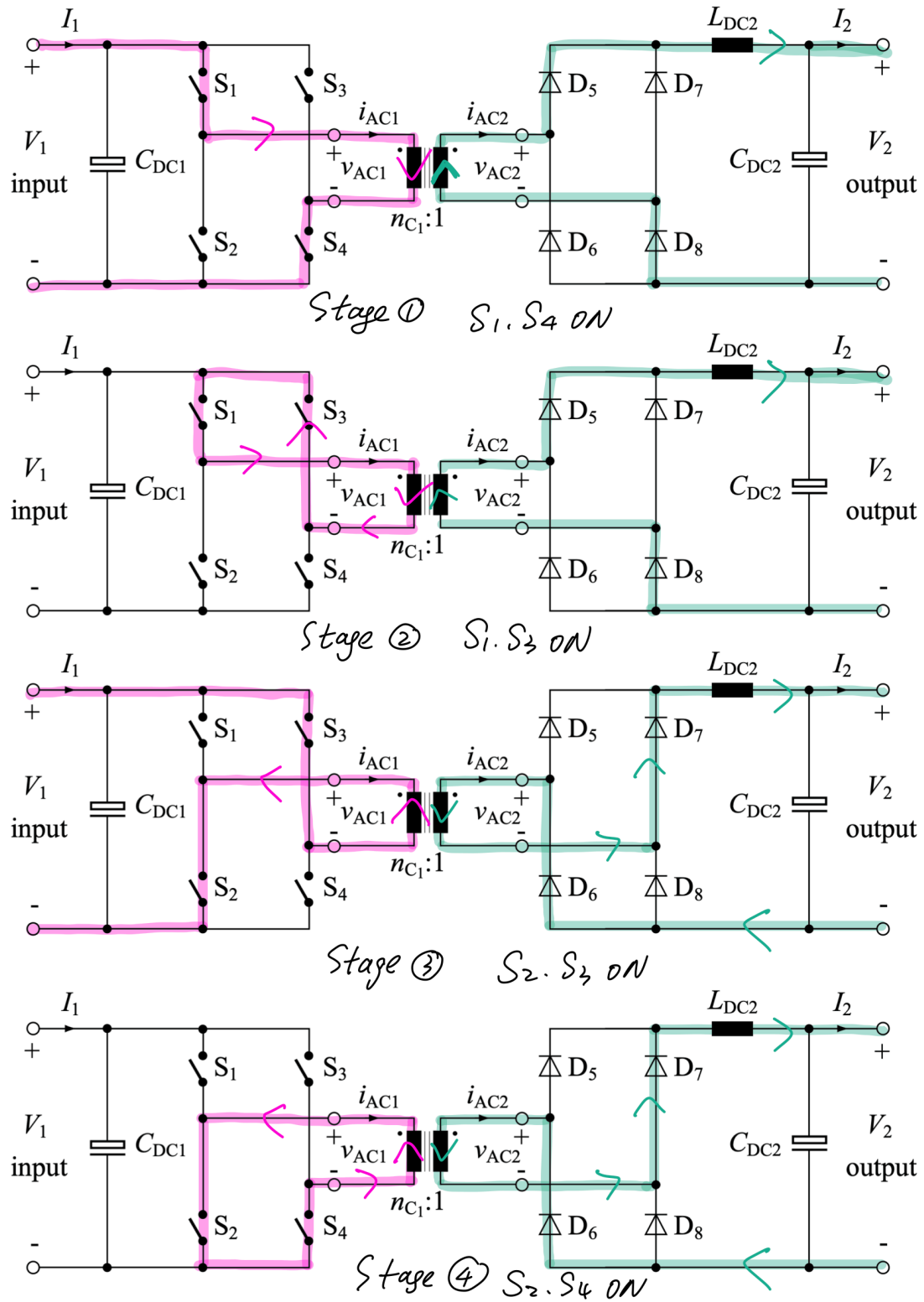


Figure 4: Operating stages of the PSFB

3: The Brute-Force Bidirectional Approach

The PSFB is unidirectional. If we want it to support bidirectional power flow, the most direct idea is to **connect two PSFBs back-to-back in opposite directions** — one handling forward power (C_1) and one handling backward power (C_2), as shown in Fig. 5.

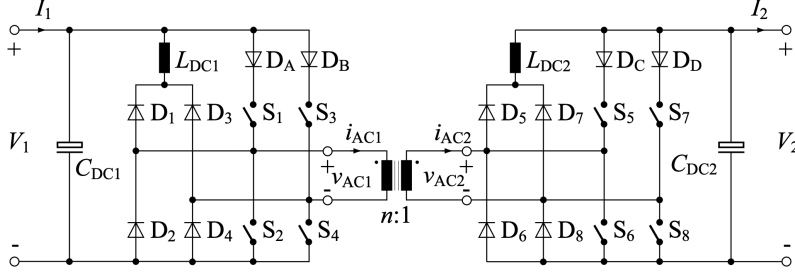


Figure 5: Bidirectional full bridge DC–DC converter topology, constructed from two unidirectional full bridge DC–DC converters

This scheme is logically guaranteed to work, but it is **extremely wasteful** in engineering terms. Counting the devices in Fig. 5, each of the two sides has:

- 4 active switches S_1 – S_4 / S_5 – S_8 (from the PSFB inverter bridge)
- 4 rectifier diodes D_1 – D_4 / D_5 – D_8 (from the PSFB rectifier bridge)
- 2 protection diodes D_A, D_B / D_C, D_D (to prevent reverse breakdown of the S devices)
- 1 DC inductor L_{DC1} / L_{DC2}

Aesthetically this circuit is very ugly, and it also suffers from a serious turns-ratio contradiction.

Forward-Mode Constraint

In forward operation the left side C_1 is the active PSFB, so applying Eq. (1):

$$V_2 = 2D \cdot \frac{V_1}{n}, \quad D \in (0, 0.5] \quad (2)$$

Since $D \leq 0.5$:

$$V_2 \leq \frac{V_1}{n} \implies n \leq \frac{V_1}{V_2} \quad (3)$$

This inequality must **hold at every operating point**. The most demanding case is V_1 minimum and V_2 maximum (then V_1/V_2 is smallest, imposing the tightest constraint on n):

$$n < \frac{V_{1,\min}}{V_{2,\max}} \quad (\text{Forward}) \quad (4)$$

Backward-Mode Constraint

In backward operation the roles swap: C_2 is the active PSFB (right side inverts, left side rectifies passively). By symmetry:

$$V_1 = 2D \cdot \frac{V_2}{n_{C_2}}, \quad n_{C_2} \leq \frac{V_2}{V_1} \quad (5)$$

Note that this turns ratio is *seen from the perspective of C_2* , secondary/primary = $1/n_{C_2}$. If the two directions share the same physical transformer, let the global turns ratio be $n = N_1/N_2$, so that $n_{C_2} = 1/n$, giving:

$$\frac{1}{n} \leq \frac{V_2}{V_1} \implies n \geq \frac{V_1}{V_2} \quad (6)$$

Again inserting the most demanding case (V_1 maximum, V_2 minimum):

$$\boxed{n > \frac{V_{1,\max}}{V_{2,\min}} \quad (\text{Backward})} \quad (7)$$

Placing Eq. (4) and Eq. (7) together:

$$\frac{V_{1,\max}}{V_{2,\min}} < n < \frac{V_{1,\min}}{V_{2,\max}} \quad (8)$$

This interval exists only if:

$$\frac{V_{1,\max}}{V_{2,\min}} < \frac{V_{1,\min}}{V_{2,\max}} \iff V_{1,\max} \cdot V_{2,\max} < V_{1,\min} \cdot V_{2,\min} \quad (9)$$

But $V_{1,\max} \geq V_{1,\min}$ and $V_{2,\max} \geq V_{2,\min}$, so the left side $\geq$ the right side. **Therefore this interval is always the empty set — for any wide voltage range there is no n that simultaneously satisfies the requirements of both directions.**

A buck topology can only step down. In the forward direction $V_1 \rightarrow V_2$ is a step-down, requiring n to point one way; in the backward direction $V_2 \rightarrow V_1$ can also only step down via buck, which means the transformer must be **used in reverse, with the turns ratio pointing the opposite way**.

Both directions demand a “step-down”, but the two directions of a single transformer must inevitably be one step-up and one step-down — physically irreconcilable.

4: SR-PSFB

Replace all rectifier diodes in Fig. 5 with active MOSFETs (the MOSFET’s built-in body diode replaces the original rectifier diode, and the 4 protection diodes D_A – D_D are no longer needed).

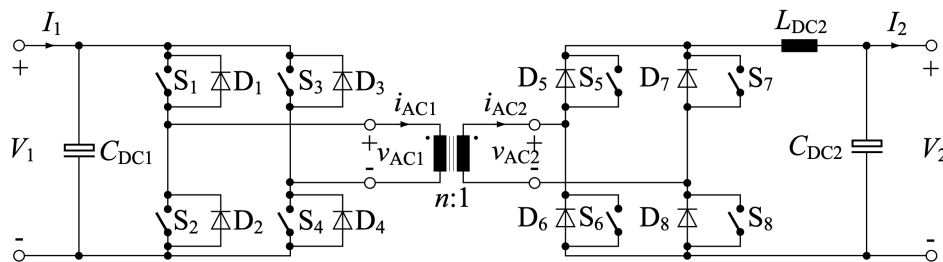


Figure 6: Bidirectional full bridge DC–DC converter (SR-PSFB)

Every time a passive device is replaced by an active one, we gain a new control degree of freedom. For instance, we can turn *all* 4 switches ON to *actively short* the transformer secondary. This capability is the key to the boost magnetization discussed below.

Forward Mode: Unchanged

In forward operation the SR-PSFB is exactly the same as the original PSFB — the left side inverts actively, the 4 MOSFETs on the right are all OFF, and the current rectifies through their body diodes (or, more sophisticatedly, via synchronous rectification). The power formula is still $V_2 = 2D \cdot V_1/n$, so **the forward mode solves no new problem**.

The advance of the SR-PSFB is therefore in the **backward mode**.

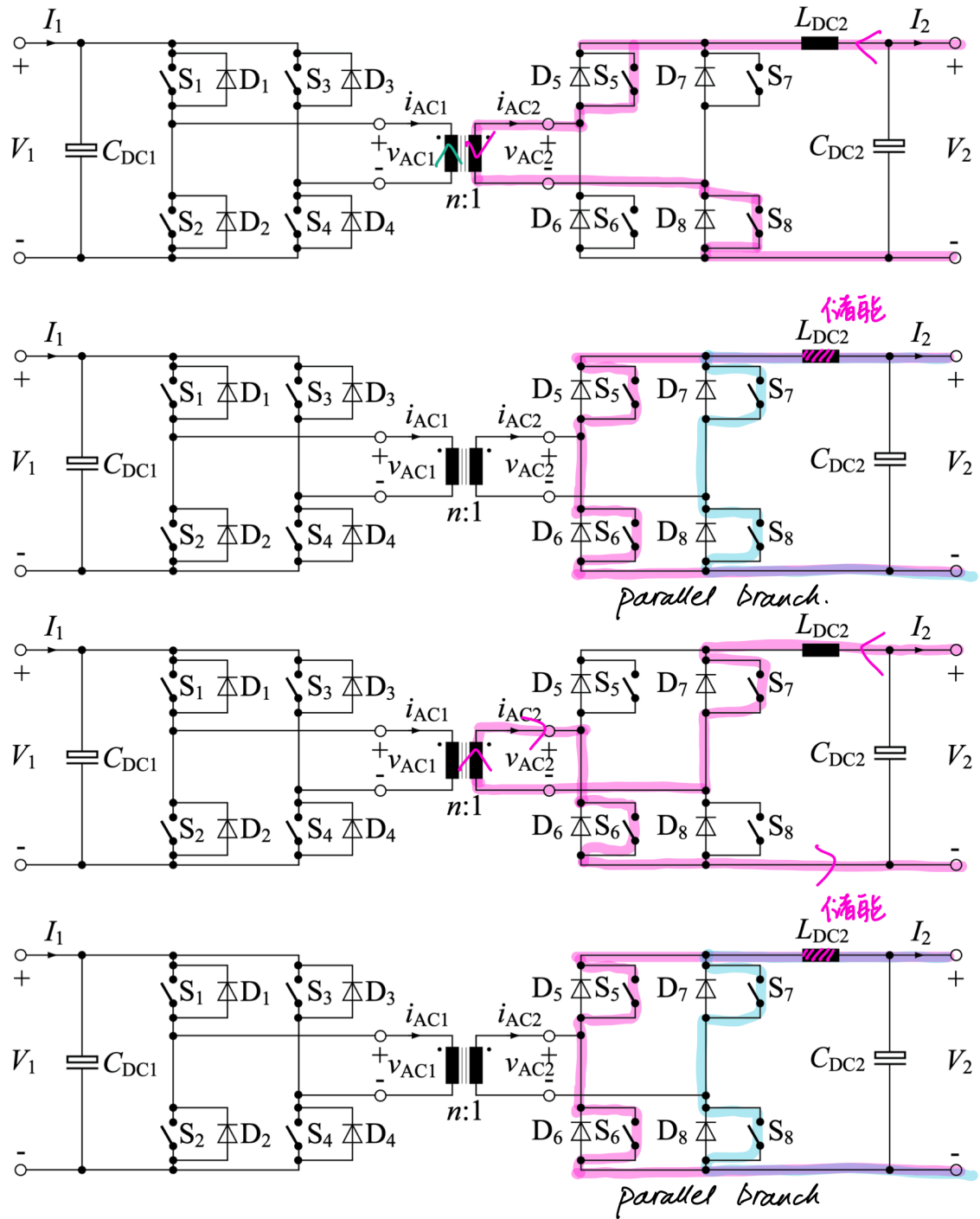


Figure 7: Operating stages of the SR-PSFB

Backward Mode: The Boost Magnetization Mechanism

Fig. 8 shows the waveforms of backward operation:

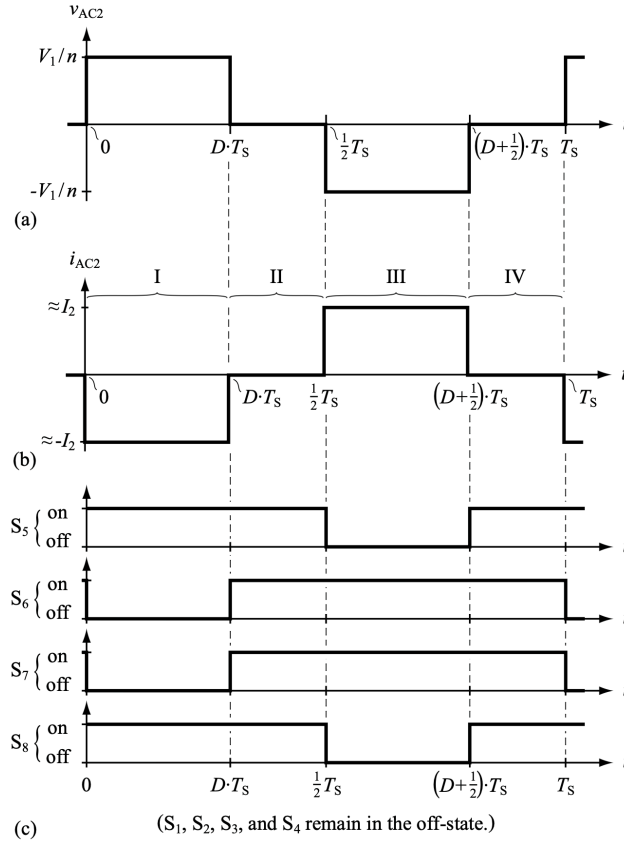


Figure 8: (a) Transformer voltage $v_{AC1}(t)$; (b) the respective switching states

In reverse operation:

Interval	Switches ON	Physical action
I	S_5, S_8 diagonal	drive $+v_{AC2}$; L_{DC2} releases energy to V_1
II	S_5, S_6, S_7, S_8 all ON	magnetize L_{DC2} (boost)
III	S_6, S_7 diagonal (reverse)	drive $-v_{AC2}$; L_{DC2} releases energy to V_1
IV	S_5, S_6, S_7, S_8 all ON	magnetize L_{DC2} (boost)

L_{DC2} is not inside the right half-bridge; it is in series between V_2 and the top rail of the right half-bridge (just as, in a boost circuit, the inductor sits in series between the source and the switch).

Why the Boost Mode Resolves the Turns-Ratio Contradiction

Recall the essence of the contradiction: a buck-derived bidirectional scheme requires both directions to be buck step-downs, leading to contradictory turns-ratio requirements.

How does the SR-PSFB break this? **Reverse operation is no longer a “buck used in reverse” but a genuine boost step-up!**

- Forward: $V_2 = 2D \cdot V_1/n$ — standard buck step-down.
- Backward: $V_1 = V_2 \cdot n/(2D)$ — standard boost step-up.

Crucially, both relations are compatible with a *single* turns ratio n : the boost inequality points **the same way** as the forward constraint, $n \leq V_{1,\min}/V_{2,\max}$, so **that one forward inequality is sufficient**, and the reverse direction adapts to different voltage ratios by adjusting D .

The turns-ratio contradiction is resolved, but **the SR-PSFB leaves behind three new problems**:

1. Forward is a buck step-down mechanism and backward is a boost step-up mechanism — **one topology, two entirely different operating mechanisms**. The controller must implement two control laws, and cannot transition seamlessly when switching power direction.
2. **Complex switching timing**: forward uses phase-shift (each switch 50%, diagonals staggered); backward uses “diagonal conduction + all-ON magnetization” (each switch with $1 - D$ duty). Two separate timing laws.
3. L_{DC2} **is still a bulky hardware component**: the volume, weight, and cost of the DC-side inductor cannot be ignored.

We need a truly “symmetric bidirectional” topology — one where both directions use the same operating mechanism and the same control law. This is the motivation for the DAB.

5: DAB

Circuit Structure

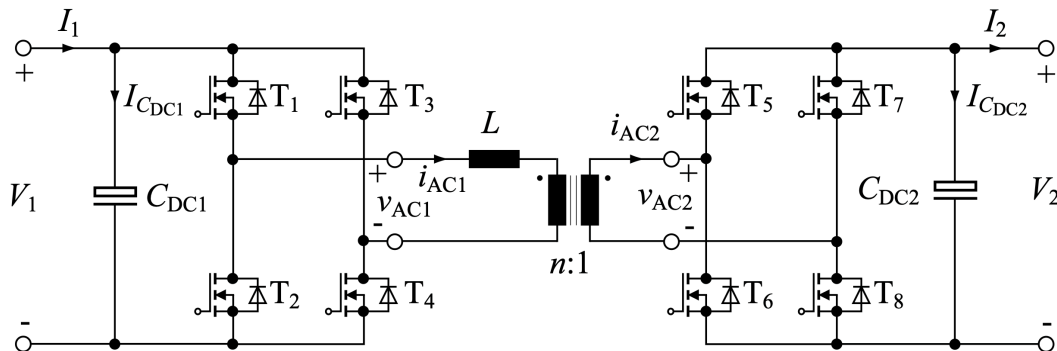


Figure 9: Dual Active Bridge (DAB) converter topology

Composition of the DAB:

- C_{DC1} : port 1 filter capacitor
- **Left full bridge** (T_1, T_2, T_3, T_4): 4 MOSFETs producing the square wave v_{AC1}
- L + **transformer**: series inductor + HF transformer ($n : 1$) — **the core energy-transfer element**
- **Right full bridge** (T_5, T_6, T_7, T_8): 4 MOSFETs producing the square wave v_{AC2}
- C_{DC2} : port 2 filter capacitor

The **only hardware difference** between the DAB and the SR-PSFB (Fig. 6) is that the DC-side inductor of the SR-PSFB is moved onto the transformer’s series line.

SR-PSFB (inductor on the DC side): it averages a pulsating DC into a constant DC — it is a *filter*.

DAB (inductor on the AC side):

- Left end of L : v_{AC1} (square wave, $\pm V_1$)
- Right end of L : transformer primary voltage = v_{AC2}/n (square wave, $\pm V_2/n$)
- Voltage across the two ends: $v_{AC1} - v_{AC2}/n$ (**the difference of two independent square waves**, which can be $V_1 + V_2/n$, $V_1 - V_2/n$, $-V_1 + V_2/n$, etc.)
- Role: **it sustains the difference of two independent voltage sources and generates a current** — it is an *energy-transfer element*.

The AC-Link Mechanism

The essence of the DAB — the synchronous-machine / transmission-line model
Considering only the fundamental component of the square waves, the DAB is simply this circuit:

$$V_1 \angle 0^\circ \longrightarrow j\omega L \longrightarrow V_2 \angle -\varphi$$

Two sinusoidal voltage sources with an inductor in between — exactly the textbook “two-machine system” or “transmission-line power-transfer model”.

Its power-transfer formula (fundamental / first-harmonic approximation):

$$P_o = \frac{V_1 \cdot V_2}{\omega L} \sin \varphi \quad (10)$$

This is precisely the power-angle equation of synchronous machines / power systems:

$$P = \frac{|E_1||E_2|}{X} \sin \delta \quad (11)$$

Characteristics of the DAB

Krismer’s thesis describes the core advantages of the DAB as follows: the main advantages are its small number of passive components, the evenly shared currents across the switches, and its inherent soft-switching properties.

- **Few passive components:** just 1 inductor + 1 transformer (one can let L be the transformer leakage inductance, sparing even the external inductor).
- **Evenly shared currents:** the 8 switches share the current stress evenly, with no distinction between “main” and “auxiliary” switches.
- **Soft switching:** all 8 MOSFETs can achieve ZVS (within a reasonable operating region).

Drawbacks of the DAB

Krismer also notes that the transformer current waveforms $i_{AC1}(t)$ and $i_{AC2}(t)$ depend strongly on the actual operating point, and that for certain operating points very high transformer RMS currents result.

When $V_1 \neq nV_2$ (**voltage mismatch**), the transformer current develops a large amount of reactive circulating current that transfers no real power but needlessly increases the RMS stress. **This is a deficiency of SPS (Single Phase Shift) modulation:** it has only one control variable φ , and cannot simultaneously optimize “power” and “low RMS”.

This is exactly why Krismer later devotes the whole of Chapter 5 to **Optimal Modulation** (EPS/DPS/TPS, multi-degree-of-freedom modulation) — a smarter modulation strategy is needed to make up for the shortcomings of SPS.

From PSFB to DAB

Topology	Innovation	Problem solved	Problem left behind
Pseudo-Resonant FB	L, C resonance at the commutation instant	Partial soft switching	Leakage L_l fights diode reverse recovery; frequency limited
PSFB	Move the DC filter inductor to the AC side; use phase-shift control	Merges leakage with reverse recovery; fixed-frequency operation; adjustable D control	Unidirectional, buck-derived, step-down only
Brute-force stacking	Two PSFBs sharing one transformer	Nominally bidirectional	Turns-ratio contradiction; no engineering solution
SR-PSFB	Diodes replaced by active MOSFETs; reverse uses boost mode	Resolves the turns-ratio contradiction; truly bidirectional	Control laws not unified across directions (buck + boost)
DAB	Move the DC-side L_{DC2} to the AC side as a series inductor L	Both directions use the same control law (φ); fully symmetric; the turns-ratio contradiction vanishes automatically	Large RMS when $V_1 \neq nV_2$

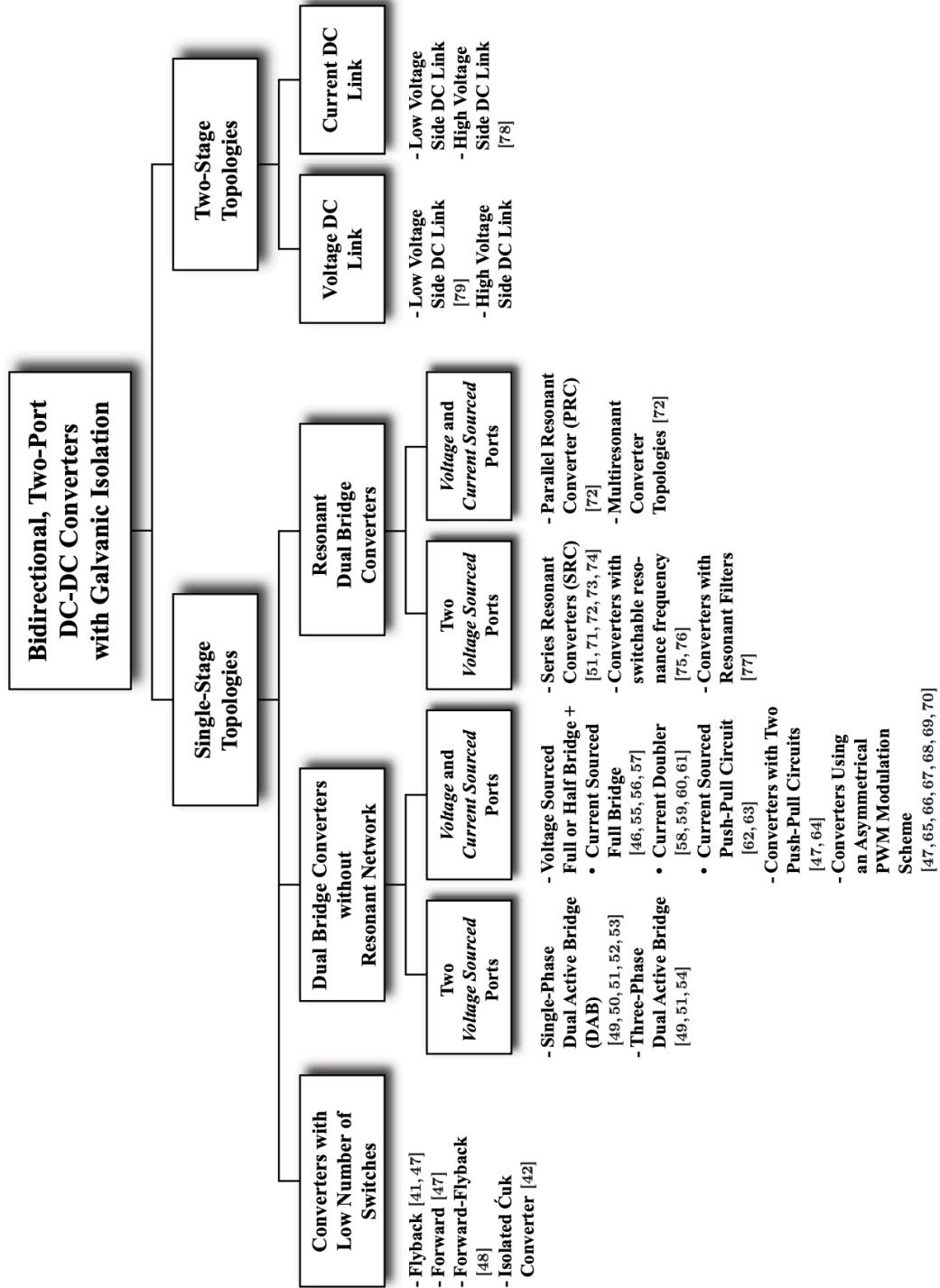


Figure 10: Topological overview of bidirectional DC–DC converters